THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET A

Chapter 14: **Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a rectangle of length 12 cm and breadth 7 cm is: 1

(a) 84 cm² **(b)** 38 cm² **(c)** 19 cm² **(d)** 91 cm²

2. The area of a triangle with base 10 cm and height 6 cm is: 1

(a) 16 cm² **(b)** 40 cm² **(c)** 60 cm² **(d)** 30 cm²

3. The area of a triangle with base 9 cm and height 12 cm is: 1

(a) 63 cm² **(b)** 108 cm² **(c)** 21 cm² **(d)** 54 cm²

4. The area of a parallelogram with base 20 cm and height 6 cm is: 1

(a) 120 cm² **(b)** 60 cm² **(c)** 52 cm² **(d)** 126 cm²

5. The area of a trapezium with parallel sides 7 cm and 11 cm and height 10 cm is: 1

(a) 97 cm² **(b)** 90 cm² **(c)** 77 cm² **(d)** 180 cm²

6. The area of a rhombus with diagonals 16 cm and 12 cm is: 1

(a) 192 cm² **(b)** 96 cm² **(c)** 28 cm² **(d)** 48 cm²

7. A rectangle has area 84 cm² and length 7 cm. Its breadth is: 1

(a) 12 cm **(b)** 14 cm **(c)** 11 cm **(d)** 77 cm

8. 1 m² equals: 1

(a) 100000 cm² **(b)** 100 cm² **(c)** 1000 cm² **(d)** 10000 cm²

9. A rectangle and a parallelogram on the same base and between the same parallels have: 1

(a) half area **(b)** equal area **(c)** different area **(d)** double area

10. The perimeter of a square is 36 cm. Its area is: 1

(a) 36 cm² **(b)** 324 cm² **(c)** 81 cm² **(d)** 144 cm²

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Write the formulas for the area of a triangle and a parallelogram. 2

12. Find the area of a rectangle 18 m by 11 m. 2

13. Why is perimeter not a measure of area? Give an example. 2

14. Find the area of a triangle with base 15 cm and height 8 cm. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A rectangular plot is 14 m by 9 m. Find its area and the cost of fencing at ₹ 30 per metre. 3

16. A rectangular plot is 20 m by 13 m. Find its area and the cost of fencing at ₹ 30 per metre. 3

17. A rectangular plot is 25 m by 8 m. Find its area and the cost of fencing at ₹ 30 per metre. 3

OR

A rectangular plot is 32 m by 15 m. Find its area and the cost of fencing at ₹ 30 per metre. 3

18. A rectangular plot is 18 m by 18 m. Find its area and the cost of fencing at ₹ 30 per metre. 3

OR

A rectangular plot is 27 m by 6 m. Find its area and the cost of fencing at ₹ 30 per metre. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Derive the area of a triangle from a rectangle. 5
(ii) Find the area of a triangle with sides 13, 14, 15 using height 12 on side 14.

OR

Case study: A farmer's field is a trapezium with parallel sides 60 m and 90 m, and the distance between them 40 m. 5
(i) Find the area.
(ii) Cost of ploughing at ₹ 2 per m².
(iii) The area in hectares (1 ha = 10000 m²).

20. (i) Derive the area of a trapezium by dividing it into two triangles. 5
(ii) Find the area of a trapezium with parallel sides 20 cm and 30 cm and height 12 cm.

OR

(i) Show that a median divides a triangle into two equal areas. 5
(ii) The area of triangle ABC is 80 cm²; D is the midpoint of BC. Find the area of triangle ABD.

END OF QUESTION PAPER — **Total: 40 Marks**

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